QSCP Hamilton Inspection Department

Inspection Plan

Site: Hamilton Plant

No.	Status	Date
QSCP-IP-ATU100 Rev 2	Issued	2010-10-15

Piping Inspection Plan - ATU-100

Scope and Objectives

This document establishes the revised inspection strategy for the ATU-100 piping circuits at the Hamilton Plant. The primary objective is to mitigate damage mechanisms identified in API RP 571, specifically focusing on internal amine corrosion and external Corrosion Under Insulation (CUI). This revision incorporates findings from recent mechanical integrity failures and updates the monitoring frequency for insulated systems.

Douglas Ferreira, as the Inspection Supervisor and API 570 inspector, is responsible for the technical oversight of this plan. The scope includes all process piping within the Amine Treating Unit, ensuring that thickness monitoring locations (CMLs) are positioned to capture the highest corrosion rates. This document QSCP-IP-ATU100 Rev 2 supersedes QSCP-IP-ATU100 Rev 1 in its entirety.

CUI Program and Mitigation Strategy

Following the issuance of QSCP-NCR-2010-007, which documented external CUI wall loss below t-min on line 4-RA-104, a mandatory CUI program is now implemented. All insulated lines operating within the CUI temperature range must undergo profile RT or have insulation windows removed every 5 years. This proactive measure is designed to identify localized external thinning before it reaches retirement thickness.

The remediation of the 4-RA-104 circuit was completed under QSCP-MOC-2010-022. This involved the replacement of the spool at CMLs -03 to -05 in August 2010. The new spool utilizes an immersion-grade coating and a revised insulation specification to prevent a recurrence of the conditions noted in QSCP-NCR-2010-007. Future inspections must verify the integrity of this coating system.

Thickness Monitoring and Interval Calculation

Corrosion rates for all ATU-100 circuits are determined by selecting the larger of the long-term rate or the short-term rate, provided the latest readings are at least 3 years apart. Remaining life is calculated as the current thickness minus t-min, divided by the established rate. The next thickness measurement is scheduled at half the remaining life, not to exceed the class maxima defined by site rules.

Class 1 circuits, including sour gas and the regenerator overhead, are limited to a 5-year thickness and external visual interval. However, the regenerator overhead circuit specifically remains on a 3-year thickness interval due to historical volatility. Class 2 amine and reflux lines are set to a 10-year thickness and 5-year visual interval, while Class 3 utility lines follow a 10-year cycle for both activities.

Responsibilities and Compliance

The QSCP Hamilton Inspection Department is responsible for executing the NDE tasks outlined in this plan. Douglas Ferreira shall review all thickness data to ensure any reading below t-min is immediately addressed via a new nonconformance report. The 138 total CMLs established in the previous revision are maintained, ensuring comprehensive coverage of all fittings and high-turbulence areas.

Compliance with this plan is mandatory for maintaining the asset integrity of the ATU-100. Any deviations from the 5-year CUI inspection window or the 3-year regenerator overhead interval must be documented and approved through the formal MOC process. This plan ensures that all piping remains within the safety limits defined by the site retirement thickness rules.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2010-10-15	_____